

CO4_AT3_INDUSTRY PROBLEM SOLVING TASK

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QUESTIONS

Q1
Smart Water Quality
Monitoring System -
30 marks

1/1 answered

Q1 Smart Water Quality Monitoring System

30 marks

Scenario / Problem Context:

A municipal water supply board is responsible for ensuring safe and clean drinking water to millions of residents across urban and rural areas. Water quality is monitored at multiple points such as reservoirs, treatment plants, and distribution pipelines. Currently, water quality parameters like pH level, turbidity, dissolved oxygen, and contamination levels are recorded manually or through isolated sensor systems. Due to the lack of a centralized data management system, the collected data is often scattered across different files and formats, making it difficult to analyze trends or detect contamination in real time. Delays in identifying poor water quality can lead to serious public health risks. Additionally, regulatory authorities require periodic reports on water quality compliance, which are currently prepared manually, consuming significant time and effort.

To address these challenges, the water board plans to implement a Smart Water Quality Monitoring System that collects, stores, analyzes, and reports water quality data efficiently. The system should support large-scale data handling, provide alerts when parameters exceed safe limits, and maintain historical records for trend analysis and decision-making.

Task Description:

Design and develop a Python-based Smart Water Quality Monitoring System using file handling and modular programming. The system should read water quality data from files or sensors, analyze it against predefined safety thresholds, generate alerts for contamination, and produce detailed reports. The application must be modular, separating data collection, analysis, and reporting functionalities for better scalability and maintainability.

Sub-tasks / Expected Deliverables:

a) Data Collection Module:

Read water quality data (location, date, pH level, turbidity, dissolved oxygen, etc.) from a CSV file and store it systematically for further processing.

b) Configuration Module:

Load safe threshold values for each water quality parameter (e.g., acceptable pH range, turbidity limits) from a JSON or configuration file.

c) Analysis Module:

Compare the collected data with threshold values and identify any deviations or contamination risks. Calculate quality status (Safe / Moderate / Unsafe) for each record.

d) Alert Generation:

Generate alerts for any parameter exceeding safe limits and store these alerts in a separate text file for immediate action.

e) Data Storage and History:

Maintain historical records of water quality data in files to enable long-term trend analysis and auditing.

f) Report Generation Module:

Generate detailed reports summarizing water quality across locations, including average values, number of unsafe readings, and overall compliance status.

g) Error Handling:

Handle missing or incorrect data such as invalid parameter values, incomplete records, or incorrect file formats using appropriate exception handling mechanisms.

h) Scalability and Extensibility:

Ensure the system can handle large datasets from multiple monitoring stations and easily support additional parameters (e.g., heavy metals, microbial contamination) in the future.

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File 1

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File 2 (optional)

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